

一、 1. A 2. A 3. B 4. B 5. C 6. D

二、 1. $\lim_{x \rightarrow 0} \frac{(1+ax^2)^{\sin x} - 1}{x^3} = 6$, 则 $a = 6$. 2. $y^2 = x$ 点 $(0,0)$ 曲率圆 $x^2 + y^2 = x$.

3. $y = \frac{x^2}{x+1}$ 斜渐近线 $y = x - 1$ 4. $f(x) = (e^x + 1)x^2$, 则 $f^{(5)}(1) = 31e$.

5. $\int \frac{1}{e^x + e^{-x}} dx = \arctan e^x + C$ 6. $\frac{d}{dx} \int_0^x \sin(x-t)^2 dt = \sin x^2$.

三、 $\lim_{x \rightarrow +\infty} \frac{x - \frac{1}{2}x^2 - \ln(1+x)}{x - \sin x} = -2$.

四、 设 $y = f(x)$ 由方程 $y - xe^y = 1$ 所确定, 求 $\left. \frac{d^2 y}{dx^2} \right|_{x=0}$.

$y' - e^y - xe^y y' = 0$, $y'|_{x=0} = e$ $y'' - e^y y' - e^y y' - xe^y (y')^2 - xe^y y'' = 0$, $y''|_{x=0} = 2e^2$,

五、 求 $y = x^2 + \frac{1}{x}$ 的极值、拐点. $y' = 2x - \frac{1}{x^2}$, $y'' = 2 + \frac{2}{x^3}$, $x_{1,2} = \frac{1}{\sqrt[3]{2}}$, -1 , 极小 $f\left(\frac{1}{\sqrt[3]{2}}\right)$, 拐点 $(-1, 0)$

六、 计算 $\int_0^2 \frac{|1-x|}{\sqrt{2-2x+x^2}} dx$.

$\int_0^2 \frac{|1-x|}{\sqrt{2-2x+x^2}} dx = \int_1^2 \frac{x-1}{\sqrt{1+(x-1)^2}} dx - \int_0^1 \frac{x-1}{\sqrt{1+(x-1)^2}} dx = \sqrt{1+(x-1)^2} \Big|_1^2 - \sqrt{1+(x-1)^2} \Big|_0^1 = 2\sqrt{2} - 2$

七、 计算 $\int \ln \left(1 + \sqrt{\frac{1+x}{x}} \right) dx$ $\left\{ = x \ln \left(1 + \sqrt{\frac{1+x}{x}} \right) + \frac{1}{2} \ln (\sqrt{x} + \sqrt{1+x}) + \frac{1}{2} x - \frac{1}{2} \sqrt{x+x^2} + C \right\}$.

原 = $\int \ln(1+t) d \frac{1}{t^2-1} = \frac{\ln(1+t)}{t^2-1} - \int \frac{1}{(t-1)(1+t)^2} dt = \frac{\ln(1+t)}{t^2-1} - \frac{1}{2} \left[\frac{1}{2} \ln \frac{t-1}{t+1} + \frac{1}{t+1} \right] + C$

八、 偶 $f(x)$ 在 $[0, \pi]$ 上连续, 证明 $\int_0^\pi x f(\cos x) dx = \frac{\pi}{2} \int_0^\pi f(\cos x) dx$, 并计算 $\int_0^\pi \frac{x \sin^3 x}{1 + \cos^2 x} dx$.

$\int_0^\pi x f(\cos x) dx \stackrel{t=\pi-x}{=} \int_0^\pi (\pi-t) f(\cos(\pi-t)) dt = \pi \int_0^\pi f(-\cos t) dt - \int_0^\pi t f(-\cos t) dt = 0$

$\int_0^\pi \frac{x \sin^3 x}{1 + \cos^2 x} dx = \frac{\pi}{2} \int_0^\pi \frac{\cos^2 x - 1}{1 + \cos^2 x} d \cos x = \frac{\pi}{2} [\cos x - 2 \arctan \cos x]_0^\pi = \frac{\pi}{2} (-2 + \pi) = \frac{\pi^2}{2} - \pi$

九、 已知 $y = f(x)$ 在 $[a, b]$ 上二阶可导, $f(c)f'(c) \geq \left[\max_{x \in [a,b]} |f(x)| \right]^2$, $c = \frac{a+b}{2}$, $b-a > 4$, 证明:

$\exists \xi \in (a, b)$, 使得 $f''(\xi) + f(\xi) = 0$

$F(x) = f^2(x) + f'^2(x)$, $|f'(\xi_1)| = \frac{|f(c) - f(a)|}{c-a} < M$, $|f'(\xi_2)| = \frac{|f(b) - f(c)|}{b-c} < M$,

$F(c) > 2M^2$, $F'(\xi) = 0$, $f(\xi) + f''(\xi) = 0$